ZP-L Illustrated Companion

The Ordinal Summit

Where the Tower Snaps to c_1

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This companion explains the ideas in plain language with diagrams and examples. It is not the formal document — every claim here restates a result already proved in ZP-L Incomputability Convergence. Consult that document for the authoritative mathematics.

What Is ZP-L Doing?

ZP-K carries a computational counterpart of the initial machine state c_0 as a KleeneStructure requirement (botCode_is_quine): a program whose behavior is determined entirely by its own index, whose existence the second recursion theorem guarantees. No equation makes c_0 that program: the identity reading is retired, and what stays a commitment is that c_0 actually runs, the occurrence commitment. ZP-L picks up the thread from a different direction: the ordinal direction. If a map ϕ assigns machine phases to ordinals — sending c_0 to everything below a certain point and c_1 above it — what is that threshold ordinal? ZP-L establishes the answer: ε_0, the first fixed point of the map $\alpha \mapsto \omega^\alpha$.

This is not an arbitrary choice. ZP-L proves that ε_0 is the minimal threshold — no monotone, tower-aligned map can snap before ε_0, and the canonical map snaps exactly there. ZP-L also establishes the connection to ZP-B: as the tower stages approach ε_0 in ordinal order, their 2-adic images converge to 0 in $\mathbb{Z}_2$, the point that is least in $\mathbb{Z}_2$'s norm preorder and greatest in its valuation preorder (z2_two_views, ZeroParadox/Valuation/Scale.lean). The same sequence witnesses both convergences simultaneously.

Real-world analogy — the unreachable horizon

Imagine a path where each step is longer than the last: you walk 1 mile, then 2, then 4, then 8, ... You are always getting further from the start, but you are also always somewhere on the path — never at the horizon. The horizon (ε_0) is the limit of all those steps combined. It exists as a mathematical object, but no finite number of steps reaches it.

The ordinal tower works the same way. The snap that ZP-E proved occurs at ε_0 is not a step you can take from inside the tower. It is the point where the tower stops and something new begins.

What Is ε_0?

ε_0 (epsilon-zero) is the smallest ordinal α satisfying the equation $\omega^\alpha = \alpha$. In other words, it is the first fixed point of the map that sends any ordinal α to ω^α (omega raised to the power α). This is not a coincidence or a definition by fiat — it is a property ε_0 satisfies and no smaller ordinal does.

The tower $0, 1, \omega, \omega^\omega, \omega^{\omega^\omega}, \ldots$ climbs through the ordinals, each stage being ω raised to the previous stage. Every stage is strictly below ε_0, and ε_0 is their limit — the supremum of the whole sequence. It is never reached from below; it is the boundary above all stages.



Each tower stage is strictly below ε-zero and maps to c0. The snap to c1 occurs exactly at ε-zero.

The tower stages 0, 1, ω, ω^ω, … all lie strictly below ε_0 and map to c_0. The snap transition to c_1 occurs at the threshold ε_0 itself. The arrow marks the snap. ZP-L §VII: epsilon_zero_snap_canonical.

> Remember: ε_0 is not "infinity." It is a specific countable ordinal — the first one satisfying $\omega^\alpha = \alpha$. Ordinals larger than ε_0 exist ($\varepsilon_0 + 1$, ε_1, and many more). ZP-L is not claiming that ε_0 is the largest ordinal or that the framework breaks down above it. It is the minimal snap threshold for monotone, tower-aligned maps — and that minimality is exactly what is proved.

Why ε₀ Is the Exact Threshold

Two facts together pin ε_0 as the minimal snap threshold. They work like a lower bound and an upper bound that meet at the same point.

Lower bound (snap cannot happen before ε_0): The fundamental sequence is cofinal in ε_0. This means: for any ordinal α below ε_0, some tower stage eventually exceeds α. A monotone map that assigns c_0 to every tower stage must therefore assign c_0 to α too (it cannot increase between a stage above α and α itself). So no ordinal strictly below ε_0 can be a snap point for such a map.

Upper bound (given hfp, the snap happens at ε_0): ε_0 is itself a fixed point of $\omega^\cdot$ ($\omega^{\varepsilon_0} = \varepsilon_0$). A map that assigns c_1 to all fixed points of $\omega^\cdot$ (the hypothesis hfp) must assign c_1 to ε_0. Together with the lower bound, ε_0 is the first place where c_1 can appear, and, under hfp, where it must; without that hypothesis a map can stay at c_0 everywhere.

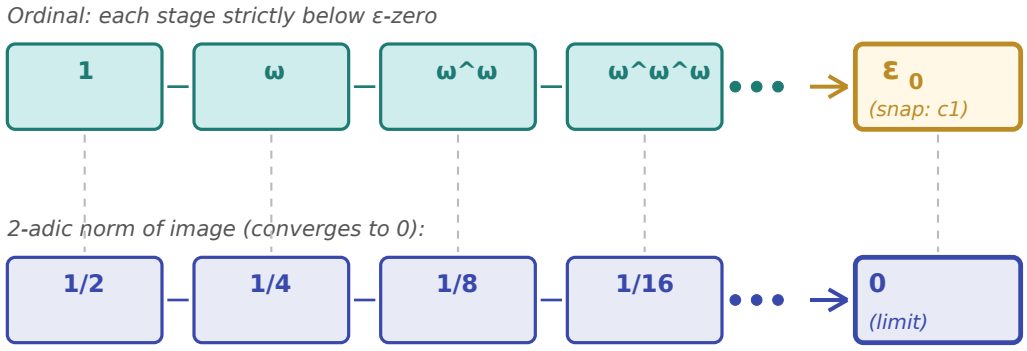
The canonical witness map $\phi(\alpha) = c_0$ if $\alpha < \varepsilon_0$, else c_1 satisfies all five conditions at once — monotonicity, tower-alignment, fixed-point-respecting, snapping at ε_0, and minimality — with no free hypotheses. This is the central result of ZP-L.

> "Minimal" does not mean "unique." ZP-L proves that no snap can occur strictly before ε_0 (under the stated conditions), not that ε_0 is the only possible snap point. Maps that also assign c_1 to ordinals above ε_0 are not ruled out. The framework fixes ε_0 as the threshold by the canonical witness, not by uniqueness of all satisfying maps.

The Two-Adic Connection

ZP-B established that 0 in the 2-adic integers $\mathbb{Z}_2$ is the valuative sink: the 2-adic valuation of 0 is infinite. So 0 is the greatest point in the valuation preorder and the least in the norm preorder, where its norm is 0 (z2_two_views). ZP-L connects this to the ordinal tower via a map, cnfToZp2, from ordinals below ε_0 into $\mathbb{Z}_2$, defined by recursion on their Cantor normal forms.

The map (cnfToZp2 in Gentzen.lean) sends the seed stage 0 to 0 itself and, from stage 1 on, the n-th tower stage to 2^n in $\mathbb{Z}_2$. Read through the valuation, the images climb: 2^n has 2-adic valuation n, rising toward the infinite valuation of 0. Read through the norm, they fall: the 2-adic norm of 2^n is $(1/2)^n$, which goes to 0 as n increases. So as the ordinal tower climbs toward ε_0 from below, its 2-adic image converges to 0 in the 2-adic metric, the point it started from. Two convergences, one tower.



Same tower, two views. Top: ordinal stages approach ε-zero. Bottom: their images' 2-adic norms approach 0.

Same tower stages, two perspectives. Teal (top row): ordinal stages 1, ω, ω^ω, … approach ε_0 from below. Indigo (bottom row): the 2-adic norms of their images 1/2, 1/4, 1/8, … converge to 0. ZP-L §VII: snap_zp2_correspondence.

This dual convergence is what snap_zp2_correspondence formalizes in ZP-L §VII: four jointly established facts about the same tower sequence — all stages below ε_0 in ordinal order, all assigned c_0 by the canonical map, their images converging to 0 in $\mathbb{Z}_2$ (from stage 1 on the images are 2^n, with 2-adic norm $(1/2)^n$; Gentzen.lean §IV), and the canonical map snapping to c_1 at ε_0.

The Kleene Connection

ZP-K carries a computational counterpart of ⊥ as a requirement of its Kleene structure: the class field botCode_is_quine, which asks the supplied code to satisfy IsComputationalQuine — a periodicity condition, eval c n = eval c (encode c + n) for every n, so the code's own Gödel number is a period of it. Kleene's second recursion theorem guarantees such a code exists. The condition does not by itself pin self-reference: the constant codes satisfy it, and they are what witnesses infinite_quine_family (ZP-K §VI). A program and ⊥ of the lattice are compared through a role rather than an identity: the Quine-atom role, a point that steps to itself and to nothing else (Motion.SelfOnlyAt), which a point of each chart occupies or provably none does. ⊥ of the lattice is its only occupant under a self-application map supplied by AbstractSelfApp (selfApp_quine_role_iff). On codes, stepping from a code to the codes whose output is what a map assigns it, no single code occupies it (evalStepMotion_not_selfOnlyAt), and for a computable map respecting equal output some class of codes up to equal output does (codes_vs_classes,

ZeroParadox/Computability/Kleene.lean). No equation across a program code and the lattice carrier is stated; they are different types, so the supplied code is not identified with ⊥, and that ⊥ actually runs stays the occurrence commitment. ZP-L identifies ε_0 as an ordinal fixed point: the first ordinal satisfying $\omega^\alpha = \alpha$, guaranteed to exist by the fixed-point theorem for normal functions. Both fixed-point proofs carry Classical.choice in their Lean formalizations; ZP-K Section IV tabulates the measured footprints. Whether these reflect the same underlying mathematical phenomenon — or are coincidentally parallel — is the open question Gentzen.lean §I poses.

ZP-L §VI (kleene_ordinal_snap_bridge) names this structural parallel explicitly. The theorem itself is purely ordinal — no Code or eval object appears in it. What the section establishes is that the hypothesis structure of the two proofs is the same: diagonalization and fixed-point existence.

Two rooms, same pattern

Imagine two mathematicians working in separate rooms. One (ZP-K) is computing with programs and asking: what program runs itself? The other (ZP-L) is computing with ordinals and asking: what ordinal satisfies $\omega^\alpha = \alpha$? Neither knows what the other is doing. When they compare notes, they find the same proof structure: the same diagonal argument, the same kind of fixed-point existence result.

ZPM maps each side into the 2-adics by its own map, the ordinal notations by cnfToZp2 and the two machine phases by snapEmbed; a bridge between the settings themselves is still open. The two rooms share a floor plan.

Convergence with Proof Theory

ZP-L works with ε_0 through ordinal fixed-point structure — ω-tower iteration, the fixed-point property of $\alpha \mapsto \omega^\alpha$, and the map cnfToZp2 from Cantor normal forms into $\mathbb{Z}_2$. Remark R-L.1 in the formal document notes a structural alignment with a result from proof theory.

Two results pin ε_0 as the proof-theoretic ordinal of Peano Arithmetic — the limit of how far PA's own proofs of transfinite induction reach — and they run in opposite directions. Gentzen reports, crediting Hilbert-Bernays, that transfinite induction up to any ordinal strictly below ε_0 is provable in PA: every ordinal short of the boundary is reached. And transfinite induction up to ε_0 itself is not provable in PA, which Gentzen proved directly in 1943: the boundary itself is not reached. A boundary needs both sides — the side that is reached and the side that is not — and neither result supplies both. A third result stands behind the pair: transfinite induction up to ε_0 is enough to prove PA consistent (1936). That is a fact about what ε_0 steps of induction prove rather than about what PA proves, so it does not on its own single out ε_0; it is also the indirect route to the unprovability half, by way of Gödel's theorem together with that 1936 result.

ZP-L and Gentzen share the ordinal itself. ZP-L defines ε_0 as Mathlib's Ordinal.epsilon 0, where ω-tower self-iteration closes on itself, and writes the ordinals below it in Cantor normal form, the notation ordinal analysis also uses (Gentzen 1943, §1). The proof-theoretic side starts from PA: ε_0 is where PA's provable cases of transfinite induction run out, the cases below it credited by Gentzen to Hilbert-Bernays (1943, p. 140, footnote 3) and the case at it Gentzen's own (1943). What is ZP-L's own is the snap: a monotone map sending the tower's stages to c_0 fires nowhere below ε_0, and its firing at ε_0 is a hypothesis, $h\varepsilon_0$ (ZeroParadox/Ordinal/Incompleteness.lean § II).

That ZP-L's snap and Gentzen's analysis both meet ε_0 is a structural observation, not an argument for either result. Both stand or fall on their own proofs.

ZP does not prove any part of Gödel's incompleteness theorems, and does not reprove Gentzen's ordinal analysis of PA. The connection is a structural observation: the same ordinal, ε_0, carries both ZP-L's snap threshold and Gentzen's analysis. Two results pin ε_0, because pinning a boundary is two-sided: it needs witnesses below the boundary and a ceiling at it, and a third result stands behind the pair. The Convergence section above states each of them, with its credit; the formal ZP-L states them in Remark R-L.1. ZP-L proves that a monotone map sending the tower's stages to c_0 fires nowhere below ε_0, and takes its firing at ε_0 as the hypothesis $h\varepsilon_0$. No formal equivalence between ZP-L's snap and Gentzen's analysis is claimed.

A Note on Classical.choice

Every theorem in ZP-L's own Lean file (Gentzen.lean) carries the axiom footprint [propext, Classical.choice, Quot.sound], inherited from Lean 4 and Mathlib rather than a novel ZP-L commitment. Results ZP-L cites from other layers are a separate question: ZP-K Section IV tabulates measured footprints on both sides.

What is notable is where Classical.choice appears, and where it does not. Across the layers ZP-L §I tabulates, the footprints are not uniform; ZP-K Section IV tabulates the measured footprints. Whether Classical.choice is structurally forced by the ZP framework — rather than merely inherited from Mathlib infrastructure — remains an open question.

Key Result — ZP-L: The Ordinal Snap Threshold

epsilon_zero_snap_canonical (Gentzen.lean §VII): The canonical map $\phi(\alpha) = c_0$ if $\alpha < \varepsilon_0$, else c_1 satisfies all five conditions simultaneously — monotone, tower-aligned (c_0 on every fundamental sequence stage), fixed-point-respecting (c_1 on every fixed point of $\omega^{\wedge}\cdot$), snapping at ε_0, and ε_0 minimal — with no free hypotheses. snap_zp2_correspondence co-proves that the same tower witnesses both the ordinal approach to ε_0 and the 2-adic convergence to 0 in $\mathbb{Z}_2$. All theorems proved without sorry. ZPM places both sides in $\mathbb{Z}_2$: cnfToZp2 sends the tower's ordinal notations there, snapEmbed sends the machine phases there, and the tower's images converge to snapEmbed's value at c_1 (zpm_triangle).